

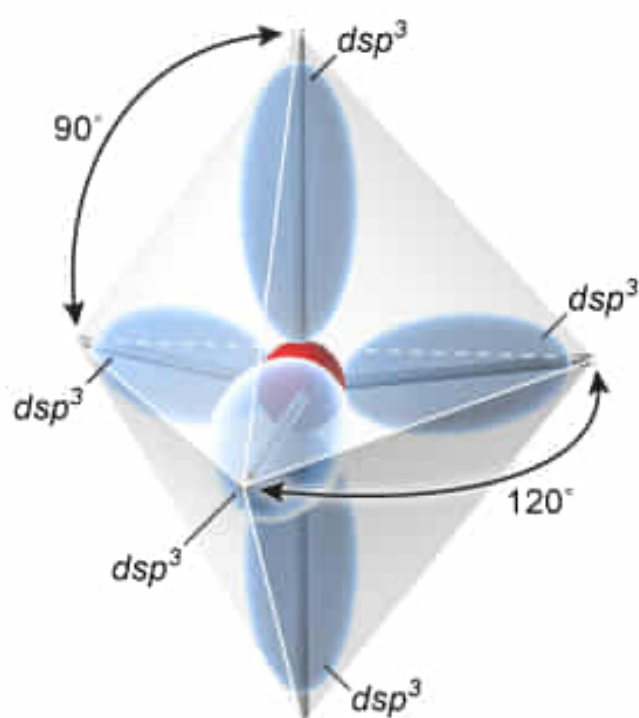
Which of the following molecular geometries could describe a compound with a central atom that is dsp^3 hybridized?

- ☐ A. Tetrahedral [9%]
- ☐ B. Octahedral [10%]
- ✓ ☒ C. Trigonal bipyramidal [74%]
- ☐ D. Square planar [6%]

Correct

74% answered correctly

Explanation:



Trigonal bipyramidal compounds
have dsp^3 hybrid orbitals

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Hybrid orbitals form when two or more **atomic orbitals** (s , p , d , f) **combine into new orbitals** of different shape and spatial orientation. An atom's **hybridization and electron geometry** are dictated by the total number of hybrid orbitals (bonding and nonbonding electron domains) formed around the atom. As the number of electron domains increases, repulsion between orbitals increases. Electrons distribute themselves into specific configurations that minimize this repulsion.

The **molecular geometry** of a compound depends on the number of bonding versus nonbonding (lone pair) electron domains. Lone pairs exert more repulsion than bonding electrons, resulting in a compression of bond angles relative to the electron geometry. As such, if two molecules have the same number of electron domains but differ in the number of lone pairs, they will have different molecular geometries.

Conservation of orbitals requires that the number of hybrid orbitals formed must equal the number of combined individual *s*, *p*, and *d* orbitals.

Accordingly, compounds with a dsp^3 hybridized central atom have five electron domains, made from one *d*, one *s*, and three *p* orbitals. To minimize repulsion, three of the hybrid orbitals lie in the same plane, 120° apart, and the other two orbitals are 90° above and below the plane. Compounds with dsp^3 hybridization and no lone pairs are **trigonal bipyramidal**. If a compound with dsp^3 hybridization has either one or two lone pairs, it will have a **see-saw** or **T-shaped geometry**, respectively.

(Choice A) Compounds with **tetrahedral** molecular geometry have four electron domains around an sp^3 hybridized central atom.

(Choices B and D) Compounds with **octahedral** (no lone pairs) and **square planar** (two lone pairs) molecular geometries have six electron domains around a d^2sp^3 hybridized central atom.

Educational objective:

Hybrid orbitals form when two or more atomic orbitals (*s*, *p*, *d*, *f*) combine into new orbitals. The total number of hybrid orbitals (bonding and nonbonding electron domains) around an atom dictates its hybridization and electron geometry whereas the molecular geometry (shape) of a molecule is determined by the number of bonding versus nonbonding electron domains.

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Interactions of Chemical Substances